Experiment 4: Image Recognition

Teaching a model to recognise what kind of document a client has sent, from just five example images per type.

WHY THIS EXPERIMENT

Experiments 1–3 worked with text; here the model learns from pictures, showing that the same "learn from labelled examples" idea works for scanned documents too.

USE CASE FOR NUVAMA

Clients send scans and phone photos of KYC forms, exchange notices, invoices and cheques; recognising the type automatically sends each to the right team and flags unreadable scans.

DATASET: "NUVAMA DOCUMENT IMAGES"

4 document types, 5 training images each (20 in total), plus 12 test images from documents never used in training: per type, 1 clean and 2 deliberately hard (blur, low light, phone-photo angle, cropped, heavy compression). Blank forms and public filings only; no personal data.

The four document types

Example	Type	Where the images came from	Who handles it at Nuvama
	**KYC** Blank KYC / demat account-opening forms (individual and non-individual)	Public blank forms from 5 different brokers and banks (Sharekhan, Motilal Oswal, JKB Financial Services, Union Bank, HSBC MF)	Onboarding & KYC team: new-account and KYC-update packs
	**Corporate action** Stock-exchange notices: dividend, bonus, buyback, split record dates	Public NSE/company filings from 5 companies (Repco Home, Ashok Leyland, eClerx, Info Edge, Remus), cropped above the signature	Corporate Actions desk: entitlements to process before the record date
	**Invoice** Business invoices, each with a different layout	FATURA dataset (CC BY 4.0), 5 different templates	Billing / accounts payable: invoices to check and pay
	**Cheque** Indian-style specimen cheques (CTS-2010 layout)	Synthetic: fictional banks, payees and numbers, marked SPECIMEN	Payments operations: cheques to verify and clear

The model: MobileNetV2 with transfer learning

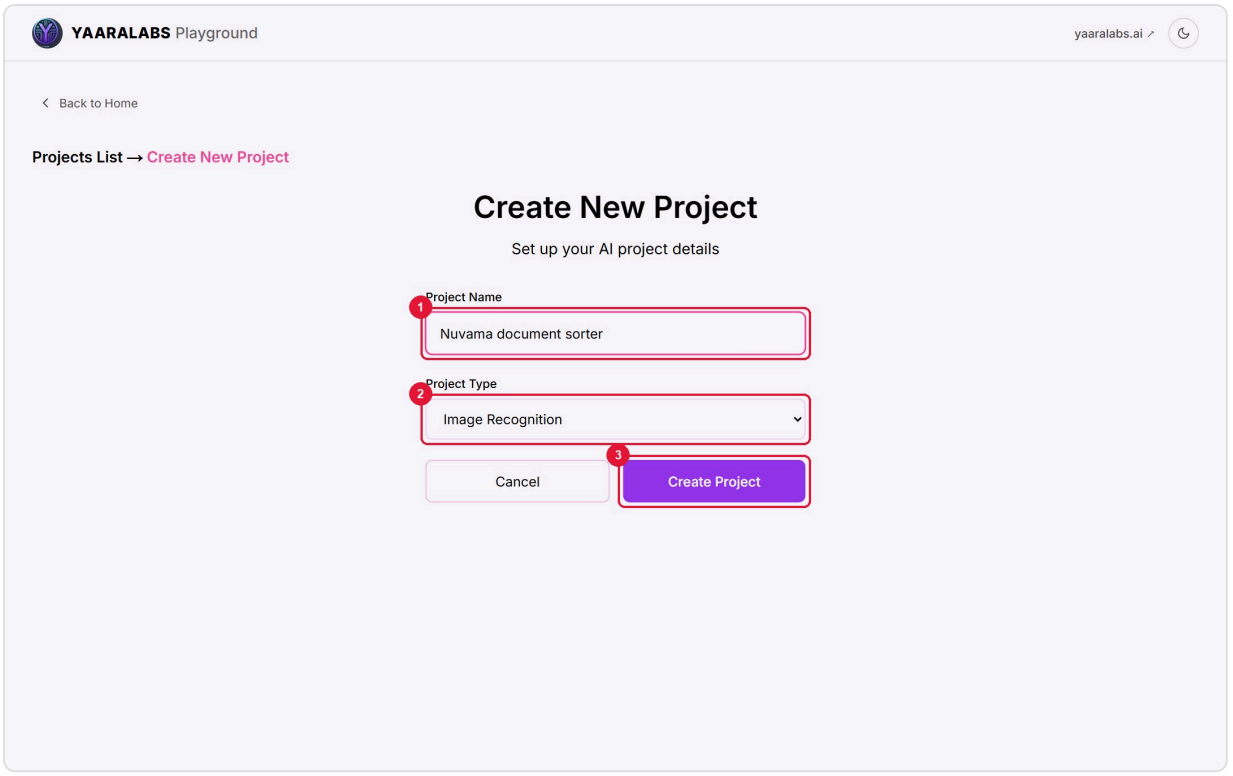
MobileNetV2 has already learned to see shapes, lines, text blocks and layouts from over a million everyday photos. We keep that knowledge and train only its last step on our 20 document images. That is why 5 examples per type are enough to learn the difference between a form, a notice, an invoice and a cheque.

Step by step

Red boxes mark what is typed or clicked; numbers show the order.

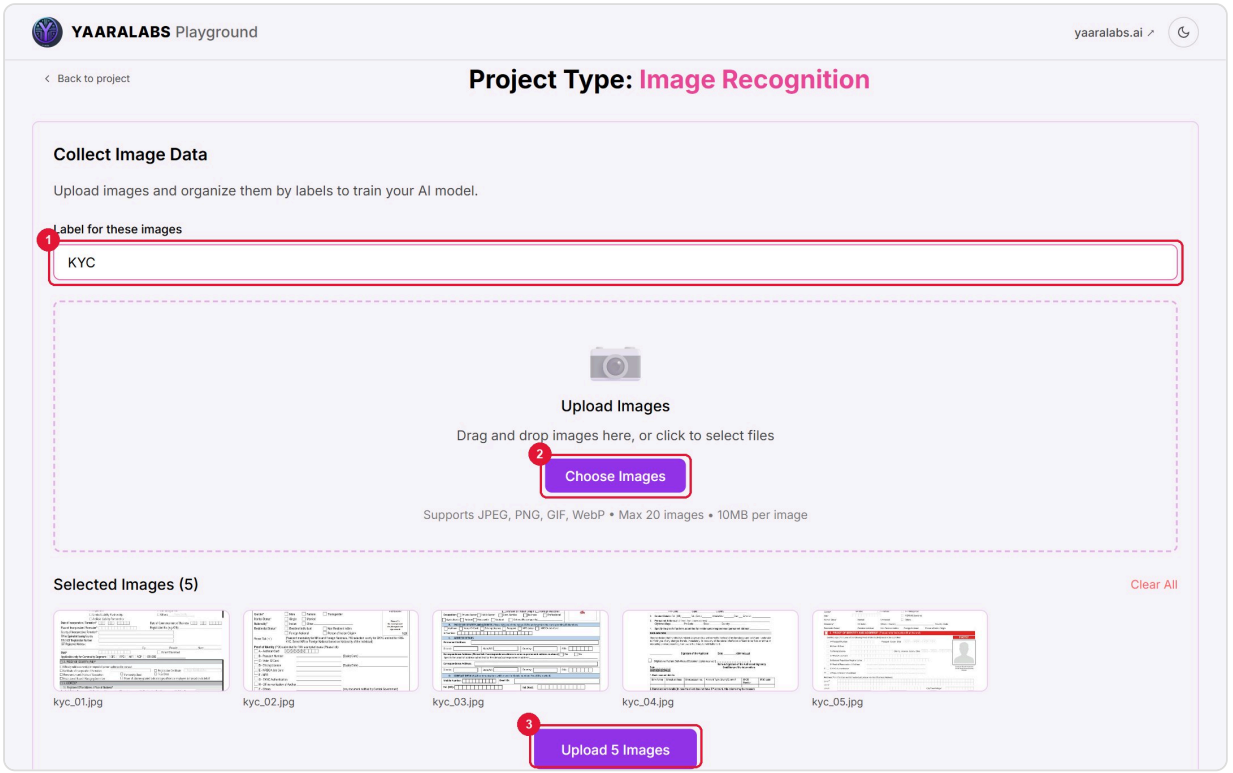
1 Create the project

Type a project name, choose “Image Recognition”, then click Create Project.



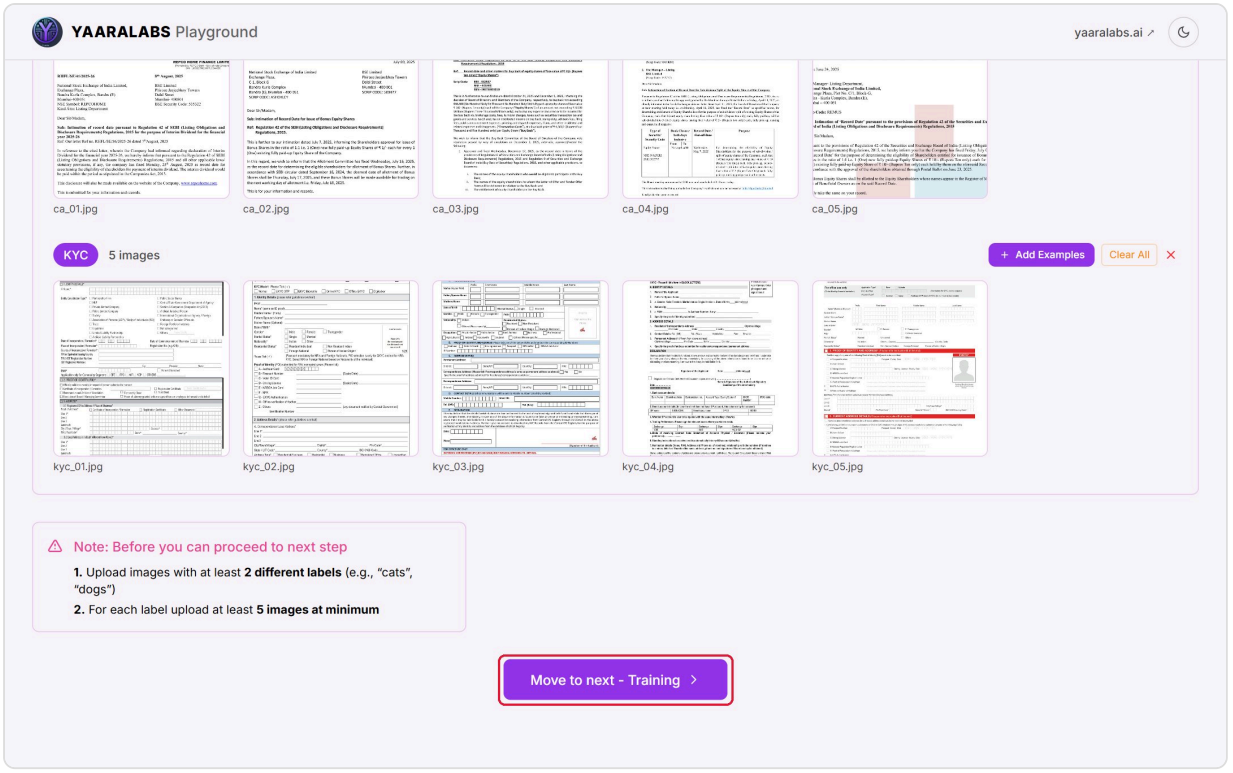
2 Add the first label and its images

Type the label (KYC), click Choose Images and pick the 5 KYC form images, then click Upload.



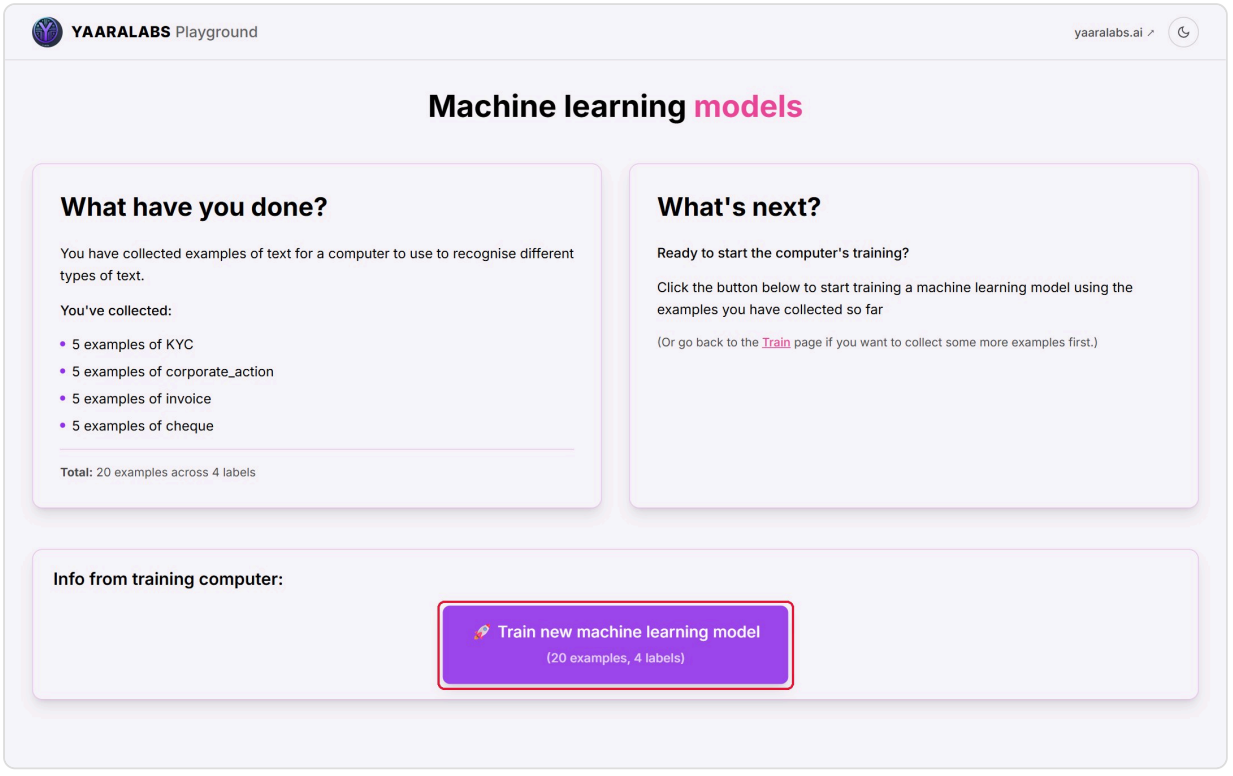
3 All four document types uploaded

Repeat for corporate_action, invoice and cheque: 4 labels × 5 images. Then Move to next – Training.



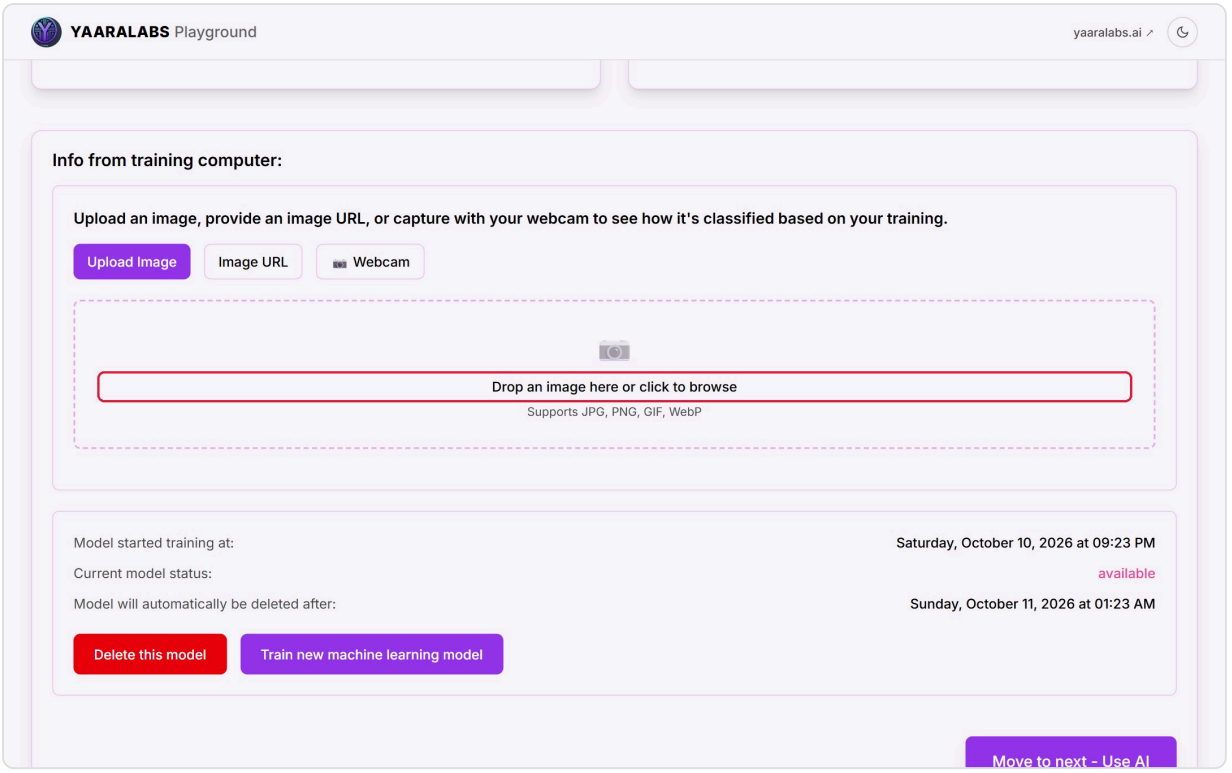
4 Train the image model

The page confirms 20 images across 4 labels. Click Train: MobileNetV2 transfer learning trains on the server.



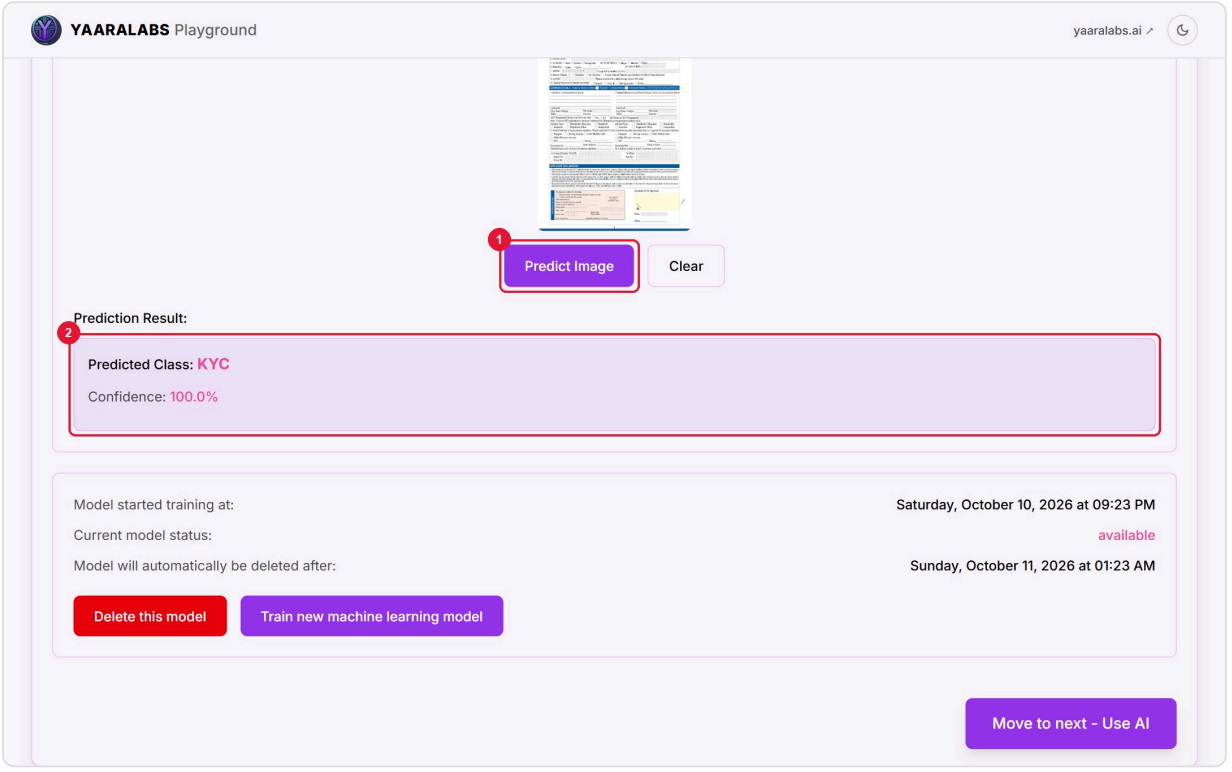
5 The model is ready

Training is done. Upload any document image here to see what the model thinks it is.



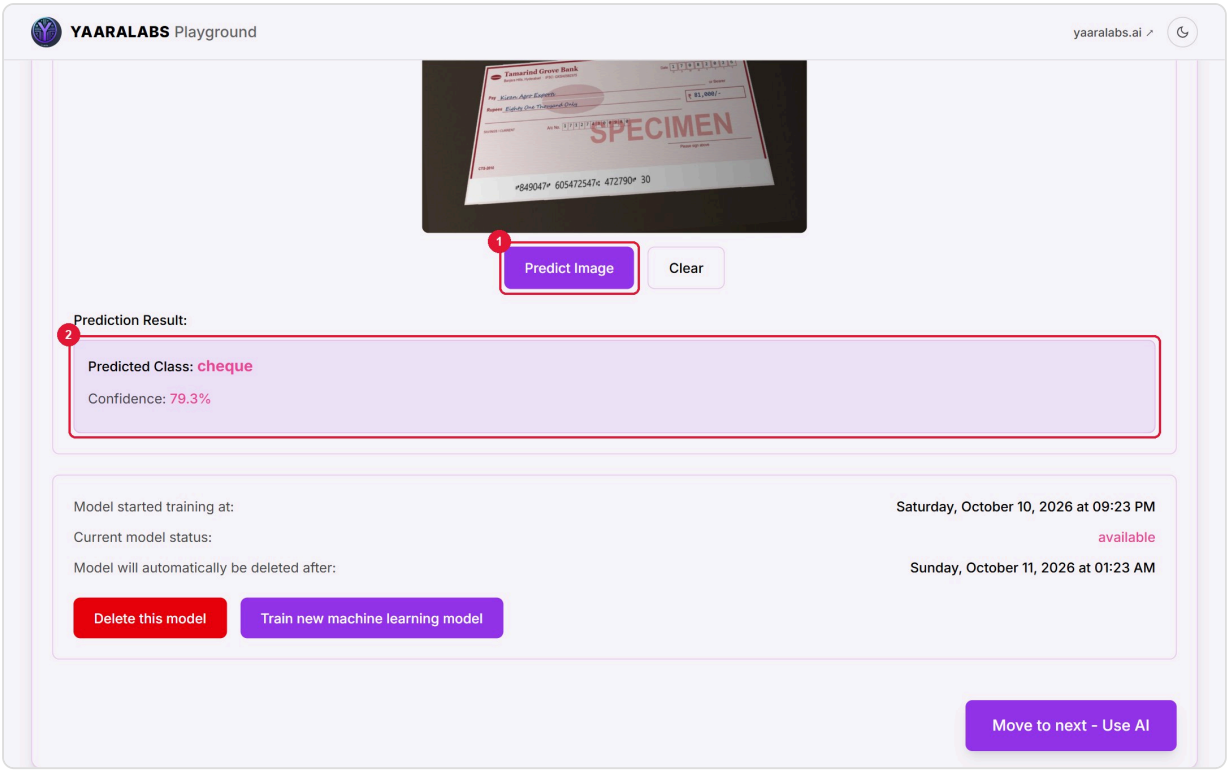
6 Test: KYC (clean)

test_01.jpg: a clean KYC image. Click Predict Image; the result shows the predicted class and confidence.



7 Test: cheque (phone-photo look)

test_05.jpg: a phone-photo look cheque image. Click Predict Image; the result shows the predicted class and confidence.



Results on the 12 test images

Overall	Clean images	Degraded images
10 of 12 correct	**4 of 4** correct	**6 of 8** correct

Test image	File and condition	Actually	Model said	Confidence	Result
	**test_01.jpg** clean: new document, upright	KYC	KYC	100.0%	**Correct**
	**test_02.jpg** clean: new document, upright	Invoice	Invoice	66.1%	**Correct**
	**test_03.jpg** partial crop: bottom 25% cut off	KYC	KYC	99.8%	**Correct**

	**test_04.jpg** clean: new document, upright	Corporate action	Corporate action	99.6%	**Correct**
	**test_05.jpg** phone-photo look: perspective skew 9 deg, rotation 4 deg, shadow gradient, on a dark desk	Cheque	Cheque	79.3%	**Correct**
	**test_06.jpg** Gaussian blur, radius 3	Corporate action	Corporate action	96.0%	**Correct**
	**test_07.jpg** heavy JPEG compression: quality 6	Invoice	Cheque	50.7%	**Wrong**
	**test_08.jpg** partial crop: right 22% cut off	Cheque	Cheque	100.0%	**Correct**
	**test_09.jpg** clean: new document, upright	Cheque	Cheque	100.0%	**Correct**
	**test_10.jpg** motion blur: 15 px horizontal	Invoice	Invoice	98.8%	**Correct**
	**test_11.jpg** low light / low contrast: brightness x0.45, contrast x0.5	Corporate action	Corporate action	71.3%	**Correct**
	**test_12.jpg** phone-photo look: perspective skew 12 deg, rotation -3 deg, shadow gradient, on a dark desk	KYC	Corporate action	69.5%	**Wrong**

Conclusion

WHAT THE EXPERIMENT SHOWS

- With only 20 training images, the model recognises document types it has never seen.
- Per type: KYC 2/3, Corporate action 3/3, Invoice 2/3, Cheque 3/3.

WHAT TO WATCH

- Mistakes: Invoice (heavy JPEG compression) read as Cheque; KYC (phone-photo look) read as Corporate action.

- Clean scans are the easy case; blur, low light, angled phone photos and crops are where it is tested.

- More and more varied training images (photos, scans, different issuers) make it sturdier.
- A low confidence score is the signal to send the document to a person.

Takeaway: transfer learning lets Nuvama build a useful document sorter from a handful of examples: 10 of 12 unseen test documents were recognised, including 6 of 8 deliberately degraded ones.